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E-commerce expands the bandwidth of entrepreneurship

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ABSTRACT

Social networks are vital for trade in developing countries, but building them is costly, creating hurdles for new entrepreneurs. E-commerce can lessen this dependence by enabling direct transactions and timely payments between producers and consumers. This article examines whether better e-commerce access boosts entrepreneurship. Using Chinese business registry data and a novel measure of county-level 3G/4G infrastructure exposure, we find that improved access significantly increases new firm entry. It also shifts the composition of entrepreneurs toward those with higher education and outside working experience, who tend to depend less on social capital in their hometowns (*JEL* L14, L26, L81, O14).

1. INTRODUCTION

In developing countries, personal trade is a defining feature due to the lack of formal institutions to enforce trade contracts. People primarily rely heavily on relational contracts to conduct trade (Greif 1993; Fafchamps 2004). Macchiavello (2022) provides a comprehensive review highlighting how, in developing countries with weak market institutions and poorly managed firms, relational contracting based on trust emerges as an alternative form of governance for economic transactions. This phenomenon is particularly prevalent when contract enforcement mechanisms are inadequate and market failures are present.

Social networks based on hometown, kinship, religion, caste, or tribe have been widely used in personal trade in developing countries due to inherently greater trust within these networks compared to those outside (Rauch 2001; Munshi 2014). In densely populated developing countries like China and India, the underlying social structure can foster specific network-based trust systems. Dai et al. (2023) discuss the presence of birth-county-based community networks in China, while Banerjee and Munshi (2004) explore the role of caste networks in India, both of which facilitate trust-based relationships in business interactions.

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However, while trust-based networks lower transaction costs within them, they come with limitations. Building and maintaining these networks can be expensive in terms of time and resources. Additionally, the exclusive nature of these networks (often referred to as "club good") makes it difficult for outsiders to enter, potentially hindering the development of talent outside the network. This restricted access can lead to a misallocation of entrepreneurial talent, where the total output could increase if the most skilled individuals outside the network replaced those on the fringes within the network (Banerjee and Munshi 2004). Therefore, network-based personal trade, despite its advantages, remains a second-best response to imperfect institutions. In the long run, as argued in North (1990) and Nunn (2007), building sound institutions is a necessary condition for transforming personal trade to impersonal trade and then further into a more impersonal and inclusive system.

While strong institutions are crucial for fostering impersonal trade in the long run, this article explores how technology, such as e-commerce, can act as a catalyst in the meantime. In the past decade, e-commerce has taken off in many countries, especially in China. In 2013, China overtook the United States as the largest e-commerce market in the world, with online sales accounting for more than 10% of retail sales, compared with 7% in the United States (*The Economist* 2013).

E-commerce has transformed personal trade into impersonal trade, devaluing social networks in at least two ways. First, it has considerably reduced the number of intermediaries between producers and consumers. Traditional trade goes through multiple intermediaries, from producers to consumers, involving a significant amount of fixed trade costs and working capital (Ahn et al. 2011; Grant and Startz 2022). Traders often use trade credit to ease working capital constraints, an endemic problem in developing countries. However, it takes a long time for traders to establish trust before extending trade credit to each other. E-commerce facilitates information flow between buyers and sellers, reducing the search cost. Online shops in production clusters could directly meet with end consumers on e-commerce platforms. This disintermediation weakens the role of social networks in long-distance trade.

Second, e-commerce has mitigated the delayed payment problems that plague traditional trade, especially long-distance trade. For example, Alibaba has set up Alipay as an escrow account to certify trade and process payments (Li et al. 2020). Due to this, sellers can quickly receive payment from Alipay after it certifies that a transaction is successful. With the speedy payment, trade credit is much less needed than before, thereby diluting the value of the social networks underlying trade credit.

Besides financial constraints, organizational barriers are common impediments to trading and entrepreneurial activities in developing countries (Nunn and Trefler 2014; Atkin et al. 2017; Atkin and Khandelwal 2020; Macedoni and Xu 2022). The supporting sectors of e-commerce, such as IT, business services, and express delivery (Goldfarb and Tucker 2019), enable entrepreneurs to overcome organizational barriers, allowing them to start and run online shops at a lower cost than a physical store. This combination of reduced reliance on social networks and trade credit, along with lower operational costs through e-commerce, leads us to hypothesize that e-commerce opens space for many potential entrepreneurs, particularly those who previously lacked social networks.

In this article, based on an in-depth case study in Baigou, a large suitcase/bag cluster and a leading "Taobao Towns" (towns known for their concentration of online businesses selling on the Taobao e-commerce platform) in China, we first examine how e-commerce penetration influences the evolution and interaction of different business models. We show that e-commerce platforms significantly reduce the search cost between sellers and consumers and enable faster payment. These features are particularly advantageous in developing countries where weak contract enforcement institutions and other market inefficiencies might hinder

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transactions between strangers (Macchiavello 2022). E-commerce reduces the need for entrepreneurs to rely solely on social capital and trust-based relationships to overcome market frictions. Additionally, the expansion of e-commerce creates new business opportunities in supporting industries, such as courier services, IT, designers, and marketing agencies.

Building on insights from the case study, the empirical section investigates whether e-commerce access translates into increased entrepreneurial activity and new business opportunities. We measure new entrepreneurs as the number of newly established enterprises in a given county and 1-digit industry from the Business Registry Data, sourced from the State Administration of Industry and Commerce (SAIC). To measure a county's access to e-commerce, we develop a spatially weighted index—exposure to 3G/4G networks—as the inverse-distance-weighted sum of the logarithm of the number of 3G/4G towers in cities beyond a 100-km radius from each focal county. This metric serves as a proxy for e-commerce access because the widespread adoption of 3G/4G technology significantly boosted online consumer penetration by enabling smartphone connectivity. This connection is particularly relevant in China, where most consumers engage with e-commerce through mobile applications rather than cable connections, thereby shifting demand toward online platforms.

To validate the proxy, we correlate our index with Alibaba's Online Business Index (OBI), which incorporates per-capita online retailers and the share of large retailers (annual transactions $>$ 240,000 yuan). After establishing the proxy's validity, we proceed in three steps: First, we test whether greater 3G/4G network exposure increases new firm entry, when controlling for county–sector fixed effects and year fixed effects. Second, we demonstrate that social and physical capital requirements are lower in counties with high 3G/4G network exposure. Finally, we examine heterogeneous effects across business types.

To characterize entrepreneur composition, we use the primarily conducted Enterprise Survey for Innovation and Entrepreneurship (ESIEC),[1] a random sample drawn from the Business Registration Data. We find that greater e-commerce access is associated with entrepreneurs having higher high-school completion rates and more professional experience, especially from outside their home regions, when controlling for county–sector fixed effects and year fixed effects. E-commerce thus reshapes entrepreneurship in two dimensions. Across sectors, firm entry increases significantly more in e-commerce-intensive sectors than in traditional sectors. Within sectors, we observe a compositional shift toward more educated and externally experienced entrepreneurs who rely less on local social capital.

Our article contributes to the burgeoning body of economics literature on e-commerce (Lucking-Reiley and Spulber 2001; Goldmanis et al. 2010; Fan et al. 2016; Lendle et al. 2016; Fan et al. 2018; Huang et al. 2018; Luo et al. 2019; Newberry and Zhou 2019; Couture et al. 2021; Startz 2021; Grant and Startz 2022). While the existing studies cover a wide range of topics—from the welfare impact on consumers, to the functioning of online platforms, to the effect on the structure of chains of intermediation—very few studies have explored the relationship between e-commerce and entrepreneurship, particularly in developing countries. By focusing on China, the largest e-commerce market in the developing world, our article helps close this knowledge gap.

Additionally, an emerging literature examines the Internet access effects on employment (Hjort and Poulsen 2019), firms' supplier access (Demir et al. 2024), and firm performance (DeStefano et al. 2018). Our article contributes to this literature by demonstrating that improved Internet access enhances online sales capacity, thereby altering both the bandwidth and composition of entrepreneurship.

[1] ESIEC is carried out by the Center for Enterprise Research, Institute of Social Science Survey of Peking University.

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Our article also speaks to the literature on the impact of entry cost on entrepreneurial activities (Blanchflower and Oswald 1998; Rajan and Zingales 1998; Dunn and Holtz-Eakin 2000; Djankov et al. 2002; Kerr and Nanda 2009; Kaplan et al. 2011; Chatterji and Seamans 2012). This body of work generally finds that lower entry costs are associated with a higher likelihood of entrepreneurial entry, particularly by those who are financially constrained. These studies often focus on the impact of institutional reforms, such as simplified firm entry regulations or financial deregulation. Our article adds to this discussion by highlighting the role of e-commerce, a key digital technology, in shaping entrepreneurial activities. By promoting division of labor, e-commerce lowers the entry barriers of starting a new business, especially in industries that rely more on external finance and relationship-specific contracts.

The remainder of this article is organized as follows. Section 2 briefly reviews the history of e-commerce and describes the inner workings of Baigou, illustrating how different business models evolve and interact as e-commerce penetrates the market. Section 3 presents an empirical analysis of the associations between e-commerce and entrepreneurship. Section 4 explores the underlying mechanisms and examines the heterogeneous effects on different types of firms, revealing how e-commerce reshapes the composition of entrepreneurship. Section 5 concludes the article.

2. BACKGROUND OF CHINA'S E-COMMERCE

2.1 Brief review of e-commerce in China

China's brick-and-mortar businesses lagged far behind those in the United States. Fortunately, the rudimentary nature of China's retail stores provided an opportunity for e-commerce to flourish. In the past two decades, e-commerce has rapidly expanded in China. Alibaba, the largest e-commerce company in the world, set up two major platforms for e-commerce, Taobao.com and Tmall.com. Taobao.com, a consumer-to-consumer platform, was established in 2003. The timing coincided with the outbreak of SARS in China. Lockdowns and quarantines make online shopping an attractive alternative for consumers restricted from physical stores. JD.com, another major e-commerce platform, was founded in 2004 in response to higher online retail demand triggered by the SARS pandemic. Alibaba set up Tmall.com in 2008 to promote business-to-consumer trade. As shown in Panel A of Figure 1, e-commerce took off the ground during this period (2004–2008). Total e-commerce turnover increased from 0.9 trillion yuan to 3.1 trillion yuan in the 5-year period, with an average annual growth rate as high as 35.9%. It is worth noting that the high growth rate is partly due to the initial low baseline for e-commerce sales in China, and e-commerce had not yet become a major retail model at that time. In 2008, e-commerce accounted for only 1.2% of total retail sales of consumer goods.

From 2009 to 2013, the annual e-commerce growth rate slowed to 26.9%. This slowdown can be attributed to several factors. At the time, online transactions had to go through online bank accounts. The application of online bank accounts was burdensome. In addition, placing orders and sending payments must go through the web through computers, which were not available to everyone. These barriers likely prevented many potential consumers from shopping online.

The year 2013 marked a milestone for fintech in China. First, the spread of smartphones and the introduction of the 3G network fueled the rapid adoption of mobile online shopping. By 2013, China had 618 million online users, with 81% accessing the Internet via smartphones (China Internet Network Information Center 2014).[2] Alibaba capitalized on

[2] http://www.cac.gov.cn/files/pdf/hlwtjbg/hlwlfzzkdctjbg033.pdf.

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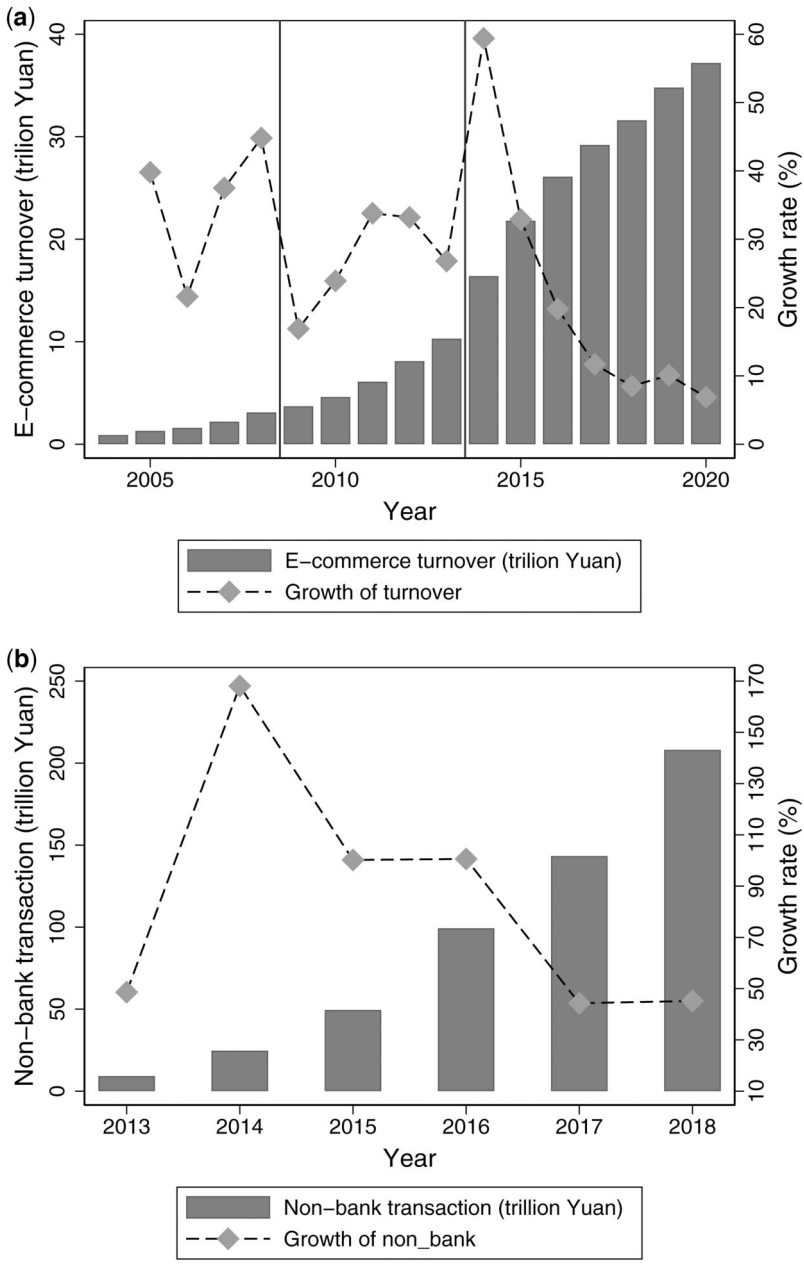


Figure 1. E-commerce turnover and nonbank transactions, by year. *Source:* Ministry of Commerce (2019, 2020).

this trend, launching its Taobao app on smartphones in 2011 and embracing the "mobile-first" strategy in 2013.[3] In November 2013, the popular annual event "Chinese Black Friday"

[3] https://www.sohu.com/a/63198168_116306.

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(November 11th) was moved to the smartphone app platform. For the first time, consumers could enjoy the convenience of smartphone purchases. It was a huge success, with one-day sales doubling year-over-year.

Second, Alibaba set up an online payment system, Alipay. In 2013, Alipay launched YuE Bao ("leftover balance" in Mandarin), a money market fund accessible primarily through mobile apps. Users can use YuE Bao to make purchases and pay bills online. The balance parked in YuE Bao earns a higher interest rate than standard savings accounts at state banks. Largely due to the launch of YuE Bao, smartphone-based payments outside the traditional bank system witnessed phenomenal growth. The number of registered Alipay mobile app users reached 100 million as of November 13, 2013, and rapidly jumped to 300 million by the end of the year, solidifying Alipay's position as the world's largest mobile payment platform.[4] As shown in panel B in Figure 1, the transaction growth rate outside the bank payment system reached 168.1% in 2014 and maintained around 100% for the following 2 years (Ministry of Commerce 2019).[5]

Fueled by the widespread adoption of smartphones and the introduction of Alipay, total e-commerce turnover in 2014 surged by 59.4%, reaching 16.4 trillion yuan, as indicated in panel A of Figure 1. The total sales of Taobao.com and Tmall.com had skyrocketed to US $500 billion by 2015, capturing a dominant 75% share of the Chinese online retail market.[6] While JD.com has experienced faster growth than Alibaba in the past several years, its market share remains significantly smaller.

However, the growth rate of e-commerce sales moderated after 2015, due to a slowdown in new Internet user acquisition. This phenomenon aligns with the concept of "regression to the mean." Initially, with a limited user base, high growth rates are natural. But as the base expands, growth inevitably slows. By 2018, Internet penetration had reached 60%, with an annual growth rate of only 3.8%. Consequently, overall e-commerce sales growth slowed, with the total value increasing from 21.8 trillion yuan in 2015 to 31.6 trillion yuan in 2018. This shift suggests a transition from extensive margin growth (increasing online consumers) to intensive margin growth (increased spending by existing users). Nevertheless, e-commerce has been the dominant retail model, accounting for 23.9% of total retail sales of consumer goods.

Overall, e-commerce penetration in China increased substantially during 2013–2017, driven primarily by the rapid spread of Internet-connected smartphones supported by 3G/4G technology. This motivates our use of cross-city variation in 3G/4G network deployment as an empirical proxy for the rise in e-commerce consumption over this period in Section 3.

2.2 Case study of a "Taobao Town": Baigou

To understand the inner workings of e-commerce, we first present a case study on Baigou, a prominent cluster in China for suitcases and bags with flourishing e-commerce activities. In 2013, Alibaba identified Baigou as one of the top "Taobao Towns" based on sales volumes on its Taobao platform. Our analysis is based on our primary surveys conducted in the region.

Baigou is located near Lake Baiyangdian, one of the largest lakes in Hebei Province, and is about a two-hour drive from Beijing. Historically, it served as a major port on the lake and

[4] https://baike.baidu.com/item/%E6%94%AF%E4%BB%98%E5%AE%9D/496859?fr=aladdin.

[5] We failed to find any data prior to 2013, perhaps because the transactions outside bank system were rare before the emergence of mobile financial app.

[6] The sales of Taobao and Tmall are available from "Internet Plus: New Engine for Chinese Economy" (23); see http://www.useit.com.cn/thread-8616-1-1.html. The data on total online sales come from "Worldwide Retail Ecommerce Sales: Emarketer's Updated Estimates and Forecast through 2019" (12), available on https://www.emarketer.com/Report/Worldwide-Retail-Ecommerce-Sales-eMarketers-Updated-Estimates-Forecast-Through-2019/2001716.

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an important market town. Over the past three decades, it has transformed into China's leading production and distribution hub for suitcases and bags. The town boasts over 3000 luggage enterprises and 10,000 family workshops, employing more than half a million local residents (Xing 2009). By 2013, Baigou housed at least 3000 online stores specializing in the sale of bags and suitcases (Alibaba Group Research Department 2013).

To deeply understand how e-commerce has transformed the real economy, we conducted a field survey in Baigou in 2014. Our research focused on three key groups: (1) traditional B2B wholesalers (for notation simplification, hereafter referred to as traditional wholesalers), (2) B2B suppliers for online shops (referred to as online suppliers), and (3) B2C online shops (referred to as online shops or online retailers). Because of their physical presence, traditional wholesale stores and suppliers for online shops could be easily spotted and interviewed. For traditional wholesalers, we used random sampling from a list provided by the market administration. Online suppliers were concentrated on two major streets. We mapped all of them before drawing a random sample for interviews.

Surveying online shops was more challenging. It was impossible to get a complete list of online shops from all platforms or the national business registry due to varying registration requirements and a lack of operational address information. For example, while JD.com and Tmall.com require registration, most Taobao shops are either registered as sole proprietorships (getihu) or are entirely unregistered informal operations (Kong et al. 2022). Online suppliers helped us locate areas where online shops were concentrated. We then mapped these key apartment buildings, randomly selected a sample, and interviewed online shop owners by knocking on every door. In total, we successfully interviewed 127 traditional wholesalers, 43 online suppliers, and 231 online shops.

2.3 Comparing the characteristics of B2C online shops and B2B wholesalers

We compared the differences in characteristics between B2C online shop owners and B2B wholesalers (encompassing both traditional wholesale stores and suppliers for online shops). Table 1 summarizes key personal background variables of the owners (e.g., age, education, and working experience) and starting capital for these two business types. There are three key findings. First, when they first started their businesses, online shop owners were noticeably younger and more educated than their counterparts in B2B wholesale businesses.

Second, B2C online shop entrepreneurs generally began with lower social and physical capital when starting their business. For instance, 78% of B2C online shop entrepreneurs were born outside Baigou, compared to only 38% of B2B wholesalers. This difference likely stems from strong social networks and reputation-based trust that are important in traditional Chinese communities. These networks give local entrepreneurs an edge over newcomers.

Regarding physical capital, opening an online shop typically requires around 29,700 yuan, covering expenses like computer equipment, rent, and basic operations. This is significantly less than the 163,500 yuan average for B2B wholesalers, suggesting that B2C online shop owners generally have less capital. Figure 2 further compares the starting capital of insiders (born in Baigou) and outsiders (born outside Baigou) across both the traditional wholesale sector and the online retail sector. The patterns are strikingly reversed across the two domains: among online shop owners, outsiders have substantially larger starting capital than insiders, which may give them a competitive edge in the emerging e-commerce market. In contrast, among traditional wholesalers, insiders (local entrepreneurs) hold higher starting capital—likely reflecting better access to traditional financing channels.

Third, B2C entrepreneurs often have more prior work experience and higher education than B2B entrepreneurs. Working as employees allows individuals to gain valuable skills that later help them spot and develop business opportunities. For example, Levine and

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Table 1. A comparison of entrepreneurial personal backgrounds: B2C online shops versus B2B wholesales.

Variables	B2C online shops	B2B wholesalers	(1)–(2) difference
Age when starting business	27.05	28.25	−1.20*
Years of education	12.46	9.35	3.12***
Outsiders (%)	77.92	37.65	40.28***
Starting capital (10,000 yuan)	2.97	16.35	−13.38***
State-sector work experience (%)	11.26	2.35	8.90***
Managerial work experience (%)	29.00	2.94	26.06***
Sales work experience (%)	12.99	4.12	8.87***
Work experience outside Baigou (%)	66.67	44.71	21.96***
Observations	231	170	

Note: Calculated by the authors based on the Baigou survey. B2B wholesalers include traditional wholesalers and suppliers for online shops. "Age when starting business" refers to the respondent's age when they started their first business. "Education" is measured in years of schooling. The variable "Outsiders" is defined as entrepreneurs whose birthplace is located outside Baigou. "Startup capital" includes the cost of building/buying/renting facilities (factories, warehouses, machines) and shops (for stall producers and online suppliers). For online shops, startup capital additionally includes the cost of establishing their online presence. "State-sector work experience" is a dummy variable indicating whether the entrepreneurs previously worked in a state-related sector (government agencies, public institutions, or state-owned enterprises) before starting their business. "Managerial work experience" is a dummy variable indicating whether the entrepreneur has prior technical or managerial experience. "Sales work experience" is a dummy variable indicating whether the entrepreneur has prior sales experience. "Work experience outside of Baigou" is a dummy variable indicating whether the entrepreneur has prior work experience outside of Baigou. The symbols * and *** stand for significance levels at the 10% and 1% level, respectively.

Rubinstein (2017) note management experience in the United States reflects strong analytical skills, which are vital for entrepreneurial success. Similarly, in China, work experience in the state sector, management, and sales can equip individuals with valuable customer knowledge and social skills. These are essential for promoting online businesses, providing customer service, and presenting products effectively.

Because online consumers rely heavily on a shop's overall design and detailed product descriptions (rather than physically examining the product), these skills directly impact how competitive an online business can be. Our data supports this: over 70% of online shop owners personally design their virtual shop web pages, and 90% manage their own product descriptions. This difference in necessary entrepreneurial skills likely explains why traditional traders find it challenging to shift into the e-commerce market.

2.4 Comparing supplier–buyer relationships between traditional and e-commerce businesses

To compare distinct business models, we examine the supplier–buyer relationships within both traditional and e-commerce businesses, starting with B2C online shops. As shown in panel (B) of Figure 3, online retailers sell bags directly to geographically dispersed consumers online, acting as the final point of sale. E-commerce platforms facilitate these transactions between strangers by reducing search costs and lowering transaction risks.

Online shops purchase bags from B2B suppliers on a spot basis, paying in cash. This results in low transaction costs, allowing them to purchase from suppliers immediately after securing online deals and minimizing the need to hold inventories. The data support this point: 60% of online retailers primarily sourcing from B2B suppliers for online shops operate with zero inventory, and 40% report never facing financial constraints due to inventory management. Consequently, online retailers can function with minimal working capital.

The use of Alipay ensures immediate receipt of funds after product delivery, eliminating the need for trade credit and minimizing the potential for disputes. Our data show that 56%

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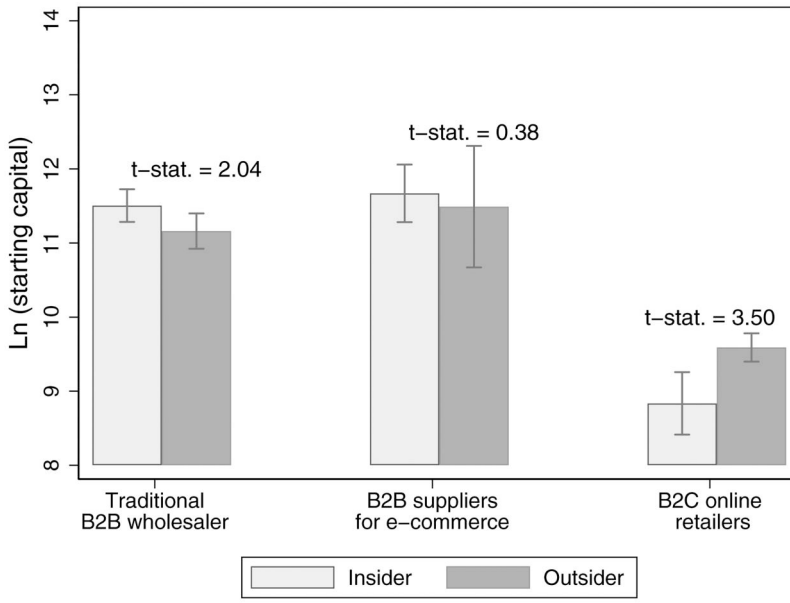


Figure 2. A comparison of startup capital for three business types by hometown status. *Source:* Based on the authors' survey data in Baigou. The labels of "insider" and "outsider" refer to the hometown status of business owners, being local or nonlocal. 95% confidence intervals corrected by the finite population correction (FPC) are plotted as vertical lines. The FPC values are 0.993, 0.899, and 1, respectively.

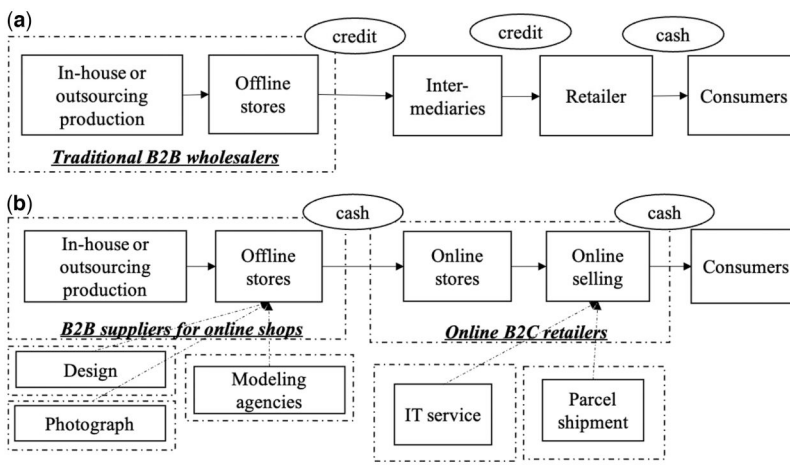


Figure 3. Traditional and e-commerce business models. *Source: Drawn by authors based on the field survey in Baigou.*

of online retailers primarily sourcing from B2B suppliers for online shops do not utilize trade credit.

Both traditional wholesalers and B2B suppliers for online shops hold similar positions within the supply chain, acting as production organizers (either self-production or

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Table 2. A comparison of client relationships: traditional wholesalers versus B2B online suppliers.

Variables	Traditional wholesalers	B2B suppliers for online shops	(1)–(2) difference
Wholesale transactions (%)	97.64	39.53	58.10***
Met at marketplaces (%)	85.04	88.37	−3.33
Number of new designs per year	46.19	164.20	−118.01***
% of payment after 1 month	33.98	12.33	21.65***
Long-term relationship (%)	69.29	32.56	36.73***
Payment dispute (%)	41.73	18.60	23.13***
Resolve dispute privately or inactively (%)	92.91	83.72	9.19*
Observations	127	43	

Note: Calculated by the authors based on the Baigou survey. "Wholesale transactions" is a dummy variable indicating whether a firm has ever engaged in wholesale deals with clients. "Met at marketplace" is a dummy variable indicating whether a firm met its oldest client through a marketplace. The calculation of "Number of new designs per year" includes 47 traditional wholesalers and 25 suppliers for online shops, all creating bag designs by themselves. "% of payment after 1 month" represents the percentage of payments received from clients 1 month or more after a transaction is complete. "Long-term relationship" is a dummy variable indicating that a firms has done business with its oldest client for more than 1 year. "Payment dispute" is a dummy variable indicating whether a firm has ever encountered payment disputes. "Resolve dispute privately or inactively" is a dummy variable indicating whether a respondent's preferred method for resolving payment disputes is to handle them privately or take no action. Finite population correction is applied in the *t*-test for traditional wholesalers and B2B suppliers for online shops, as 0.993 and 0.899, respectively. The symbols * and *** stand for significance levels at the 10% and 1% level, respectively.

outsourcing). As shown in panel (A) of Figure 3, in the traditional business, wholesalers sell bags to intermediaries (traders or middlemen) who then connect with retailers or individual customers. In contrast, the e-commerce model streamlines the process, enabling online retailers to purchase directly from suppliers and sell to end users themselves through online platforms.

While both traditional wholesalers and B2B online suppliers conduct business with customers in person and through market channels, the size and nature of their typical transactions differ significantly. For example, as shown in Row 1 of Table 2, 98% of traditional wholesalers do business with wholesaler customers, whereas only 40% B2B online suppliers sell to wholesalers.

To define relational contracting based on customer longevity, we followed the criteria established by Macchiavello and Miquel-Florensa (2017), considering a partnership to exist if it has lasted for 1 year or longer with the supplier's oldest customer. As shown in Row 5 of Table 2, 69% of traditional wholesalers have at least one such long-term relationship, compared to only 33% of B2B suppliers for online shops. Notably, 63% of traditional wholesalers have a customer relationship exceeding three years, which aligns with the threshold used in relevant literature.

The bottom two rows of Table 2 display the occurrence of payment disputes and resolution. Traditional businesses experience payment disputes more frequently, with 42% of wholesalers reporting such issues compared to only 19% in online businesses. In both sectors, private or inactive resolution is the preferred approach in both sectors, chosen by 93% in traditional business and 84% in online business.

A key difference emerges when comparing the upstream suppliers of traditional wholesalers and B2B suppliers for online shops. Beyond the common need for raw materials like leather and hardware, e-commerce specifically necessitates additional inputs. Design is a key input. Suppliers for online shops face heightened demand for new designs driven by online customers' preference for variety. Recognizing design as a core competence, 84% of B2B suppliers for online shops outsource production processes, focusing solely on design and

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Table 3. Number of firms and total registered capital by industry in Baigou.

Industry	Number of firms			Total registered capital (10,000 yuan)		
	2012	2017	Annual growth (%)	2012	2017	Annual growth (%)
Bag production and sales	883	5191	42.51	1974	12,240	44.03
IT services	68	379	41.00	20	921	114.49
Express delivery	59	206	28.41	223	841	30.39
Business services	100	774	50.57	249	3676	71.35
Other industries	1991	7498	30.37	10,847	35,791	26.97

Note: Calculated by the authors based on the business registry data. Bag production and sales are identified by their three-digit NEIC code, including bag-related production, wholesaling, and retailing. IT services, express delivery, and business services are identified by their two-digit NEIC code. All remaining firms fall under "other industries."

development. More than 40% of them source new designs from specialized design companies, thus demonstrating how e-commerce promotes the growth of the design industry.

Since online consumers cannot physically examine bags, high-quality product photos are crucial for influencing purchasing decisions. Recognizing this, B2B suppliers for online shops typically outsource product photography to professional studios. These photos are often provided free of charge to online shops, bundled with the bags themselves. Our data reveal that over 60% of online shops utilize the product images provided by their suppliers. Beyond the technical photography, the use of aesthetically pleasing models, particularly foreign models, further enhances the perceived international design and taste associated with the store. This emphasis on visual presentation contributes significantly to the growth of the fashion and culture-related entertainment industry, including photo studios and modeling agencies.

Finally, as shown in Panel (B) of Figure 3, online shops outsource IT and delivery services to other suppliers. The Internet-driven nature of e-commerce necessitates various IT services, including online customer service, Internet connectivity, and software development. The high frequency of deliveries in e-commerce necessitates a shift beyond traditional truck rental models. New delivery companies have emerged to cater to the specific needs of collecting and transporting individual parcels, further stimulating growth within this industry.

Table 3 displays the number of firms and their total registered capital in Baigou's key industries, using data from the national business registry database, comparing figures from 2012 (before the rise of e-commerce) and 2017 to calculate annual growth rates. As shown in the table, business services encompassing advertising (design), accounting, consulting, and entertainment services (photo studios and modeling agencies) experienced a remarkable annual growth rate of 51% in the number of companies and 71% in total registered capital. Similarly, Internet-related industries witnessed significant growth, with annual growth rates of 41% and 114% for the number of companies and total registered capital, respectively. The bag manufacturing industry saw an annual growth rate of 44%. By comparison, the annual growth rates for the number of firms and registered capital across all other industries were only 30% and 27%, respectively.

2.5 The transition from traditional business to e-commerce business

Approximately 30% of B2B traditional wholesalers participate in online B2B sales. Moving online requires a different approach: more frequent new product types, but in smaller volumes. This means wholesalers need to invest more in design and development. Similar to the model in Melitz (2003), only those traditional wholesalers with greater capabilities and resources can successfully make this jump.

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Traditional business typically covers a wide range of bags, including men's and women's bags, suitcases, and student bags with varying levels of quality. However, online customers, who are mostly female, prefer a diverse selection of good quality, affordable bag styles. Therefore, most traditional wholesalers who expanded online had previously focused on producing moderately priced, female-oriented bags as the additional investment required for online business is lower for them. The traditional wholesalers who broaden their business scope generally have significantly larger initial assets and more employees. On average, they have assets of 170 thousand yuan and 23.7 employees, compared to firms that remain solely traditional, which have assets of 110 thousand yuan and 11.2 employees for firms.

While starting a B2C online shop requires a relatively small initial investment, as shown in Table 1, it demands new skills for entrepreneurs. Despite 50% of traditional wholesalers expressing willingness to start an online shop, only 17% have actually done so. The main obstacle for most is a lack of IT skills, with over 40% of respondents admitting that they do not know how to use computers or set up an online shop. Attracting online consumers and implementing effective online promotion strategies also present significant challenges. Among B2B suppliers for online shops, 70% have online shops themselves. However, these primarily function as model stores for online shop owners, rather than active sales platforms.

While the Baigou case study provides valuable insights into the inner workings of traditional and e-commerce business models, it remains unclear if these observations apply to China as a whole over a longer period. To address this limitation, we formally test the relationship between e-commerce development and entrepreneurship using comprehensive administrative business registry data and primary survey data in the next two sections.

3. EMPIRICAL ANALYSIS OF THE ASSOCIATIONS BETWEEN E-COMMERCE AND ENTREPRENEURSHIP

3.1 Data description

We measure entrepreneurship in China from 2013 to 2017 by counting new businesses each year. These data come from China's official business registry, maintained by the SAIC.[7] According to China's Corporation Law, all companies must register with the SAIC before opening a business. All companies in China must register with SAIC, so this database includes every registered business, from small to large. The SAIC administrative database includes information on the starting date, four-digit industry code, location, ownership, registered capital, and legal representative and shareholders of registered businesses. If a business goes bankrupt or its license is revoked, the exit date is also recorded.

The database available to us includes the first six digits of personal IDs of legal representatives for firms registered before 2016. According to the business registry data, 90% of the largest shareholders are legal representatives as well, and 75% of legal representatives hold key positions in their firms, such as owners, managers, and directors. Thereby, we can regard legal representatives as a good proxy for entrepreneurs. The first six digits of their IDs, which indicate the location of their household registration when the ID was first issued, typically correspond to their county of birth. By comparing the hometown addresses of the legal representative with the company's

[7] In the same vein, Guzman and Stern (2020) use business registry data to study the evolving patterns of entrepreneurship in the United States. China has 20 1-digit industries based on the National Economic Classification of Industries 2017. (http://www.stats.gov.cn/tjsj/tjbz/hyflbz/201710/t20171012_1541679.html). We exclude two industries, that is, S (public administration and social organization) and T (international organization), from the sample. Considering the presence of large variations in capital requirement, production organization, and technology across the 1-digit industries, we aggregate the data to the county—1-digit industry—year level and control for the fixed effects of 1-digit industries in regressions.

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registration addresses, we can infer whether entrepreneurs are in local areas or not. In the analysis about local and nonlocal entrepreneurs, we restrict the sample period up to 2015.

We also use firms' initial registered capital as a measure of capital barriers to entry. The initial registered capital is the authorized amount of investment committed by all investors. Banks often use it as a key indicator to determine credit lines when extending loans to a firm. After the Business Registry Reform in 2014, SAIC requires firm owners to report the amount of initial capital when registering their firms, and it randomly inspects firms over the next 2 years to ensure owners have actually invested enough capital.

To assess a county's access to e-commerce, we construct a spatially weighted index: exposure to 3G/4G networks. This index is calculated as the inverse-distance-weighted sum of 3G/4G networks in cities beyond a 100-km radius from each focal county. This metric is a proxy for e-commerce access because the spread of 3G/4G diffusion significantly increased online consumer penetration by enabling smartphone connectivity. This approach highlights the crucial role of online customer accessibility from the demand side. Before the widespread adoption of smartphones and 3G/4G networks in China, online shopping was largely inaccessible for most people. As explained in Section 2.1, placing orders and making payments required Internet-connected computers, which limited participation. The introduction of 3G/4G wireless networks dramatically expanded the number of online consumers by enabling connections through smartphones. We measure the coverage of these networks by the number of existing 3G/4G towers at the city level by year. Information on the establishment year and GPS coordinates for these towers is recorded in the Global OpenCelliD Cell Tower Database.[8]

Panel A of Figure 4 shows the number of newly constructed 3G/4G towers during the period 2013–2017, with an average of 1349 towers per prefecture. To measure accessible online consumers, we leverage the aggregated number of 3G/4G towers from neighboring cities, weighted by the inverse of their distance to a central county. This weighting approach reflects transportation costs between locations, a common practice in the gravity model of trade literature. The exposure to 3G/4G networks is defined as follows:

$$Exposure_{it} = \sum_{j|distance_{ij}>100} \frac{1}{distance_{ij}} \log\left(3G/4G\ \ tower_{jt}\right) \quad (1)$$

where i denotes the focal county, and j indexes cities located at least 100 km away. Exposure is calculated as the weighted sum of the logarithm of the number of 3G/4G towers in city j, weighted by the inverse of the distance from city j to county i.

To validate the use of 3G/4G network exposure as a measure of online business, we utilize China's e-commerce index developed by Ali Research separately at the city level and rural county level for the period 2013–2017. The e-commerce index comprises two subindices: the online business index (OBI) and the online shopping index (OSI). Both are constructed as a panel data set (Luo et al. 2019). The OBI and OSI are defined as follows:

[8] It's important to acknowledge limitations in the OpenCelliD data. Since this database relies on user contributions, it may not cover all cell towers in China. For instance, in 2016, the Ministry of Industry and Information Technology (MIIT) reported 830,000 new 3G/4G towers built in China, whereas OpenCelliD only recorded 217,000 new towers. We assume this under-coverage is random across prefectures and years, after controlling for county and year fixed effects.

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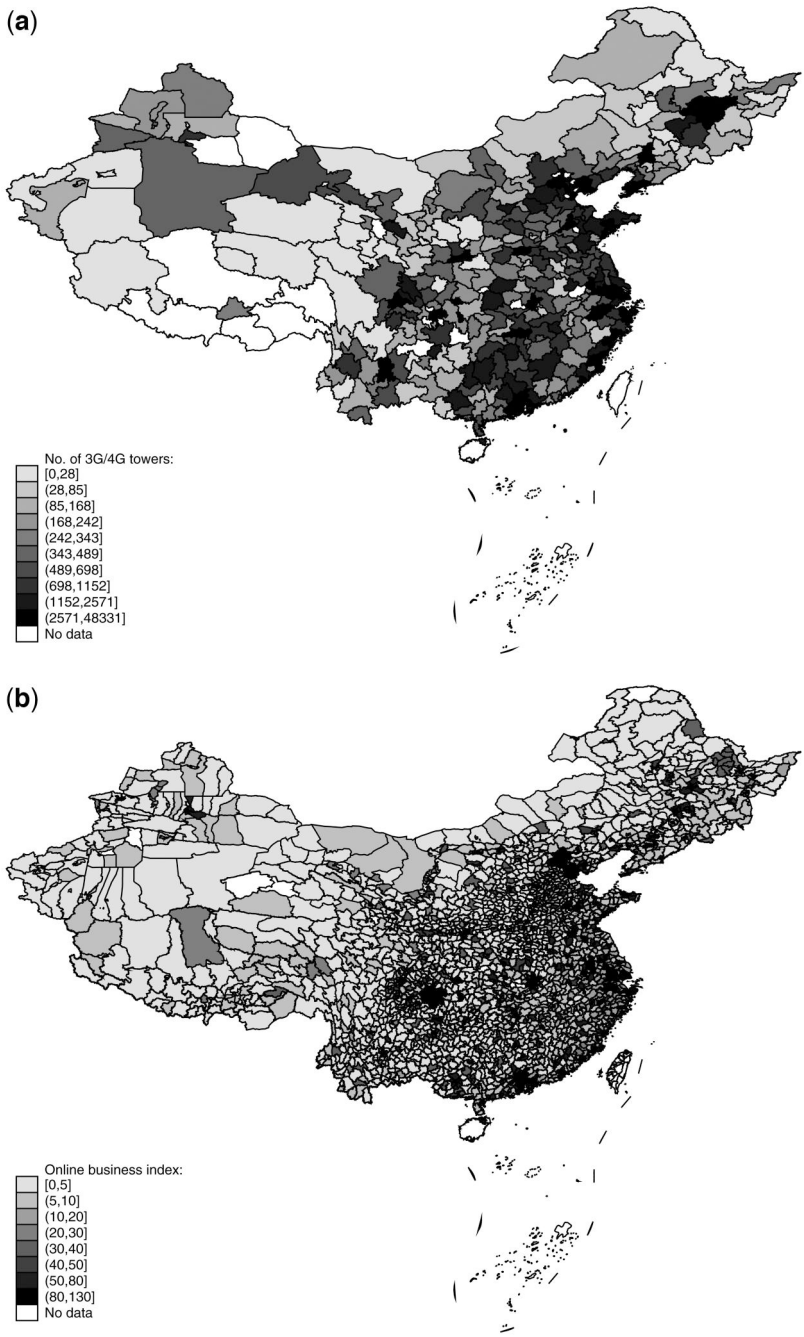


Figure 4. 3G/4G tower construction and e-commerce online business index. *Source:* Global OpenCelliD Cell Tower Database and Ali Research Institute.

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$$OBI = \left(\frac{\#\text{online shops}}{\text{population}}\right) \times 0.6 + (\%\ \text{shops with annual volume} > 240000\text{RMB}) \times 0.4$$

$$OSI = \left(\frac{\#\text{online buyers}}{\text{population}}\right) \times 0.6 + (\%\ \text{buyers with annual spend} > 10000\text{RMB}) \times 0.4$$

They are further normalized by dividing the two indexes by the maximum expected value and multiplying by 100. The maximum expected value for OBI equals 13.47% * 0.6 + 100% * 0.4. The maximum value for OSI is 63.26% * 0.6 + 100% * 0.4, when 80% of adults (79.07% of the total population) purchase online, and all of them are active online buyers. In principle, the indexes should range from 0 to 100.[9] For urban districts (same administrative level as rural counties), we use the city-level ecommerce index as a proxy, because the data at the district level is not available, and the variation in economic development across urban districts within a city is minimal. For rural counties, the e-commerce index is at the county level.

The OBI index measures a county's capacity for online sales by incorporating the percentage of online retailers per capita and the percentage of large online retailers (those with transaction volumes exceeding RMB 240,000). In contrast, the OSI primarily reflects consumer online purchasing activity, which may not directly correlate with local entrepreneurial endeavors. E-commerce is unevenly developed across space, exhibiting significant regional variation. As Panel B of Figure 4 illustrates, Alibaba's 2017 OBI clearly demonstrates this disparity. The OBI reveals that online sales are predominantly concentrated in coastal areas, often coinciding with manufacturing production centers.

Appendix Table A1 details the relationship between 3G/4G network exposure and the key indices: OBI and OSI. Assuming exposure to 3G/4G networks is exogenous, we would expect a correlation with OBI but not with OSI. The data support this expectation: exposure is positively associated with OBI (even after controlling for county and year fixed effects), but has no correlation with OSI. These findings also align with the spatial distribution patterns of exposure to 3G/4G network and OBI, as illustrated in Panels A and B of Figure 4.

The control variables used in the regression analyses include population density, log population, and the share of primary and secondary sectors at the county level. For urban districts, we use city-level statistics from the China City Statistical Yearbooks, while data for rural counties comes from the China County Statistical Yearbooks.

Table 4 summarizes all variables. We drop counties that miss control variables or the Alibaba e-commerce index, ending up with 1789 rural counties and 1002 urban districts in the sample. In total, for the period 2013–2017, this results in 13955 $(= (1789 + 1002) \times 5)$ observations. The number of firm entries is aggregated at the county and 1-digit industry level for each year over the period 2013–2017. We exclude county–industry–year cells without any firm entries throughout the entire period. However, it is still possible that some counties and industries have zero entries in specific years. The number of cells with zero values accounts for 13.8% $(= (244745 - 210949)/244745)$. Since we cannot take logarithms

[9] According to the Global Entrepreneurship Watch China Report (2002–2012) (Han 2013), 24% of the population aged 18–64 were entrepreneurs in 2011. For the whole population, the percentage would be 16.84%. If assuming 80% of new businesses are online, the percentage would be 13.47%. The maximum expected value initially assumed turned out to be conservative. Because of the rapid increase in the number of firm entries following the registration reform in 2014, the index exceeds 100 in a few cities after 2014. Since the index is only used as a scalar, the empirical results are not sensitive to the choice of the maximum expected value.

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Table 4. Summary statistics for key variables used in regressions.

	(1) OBS.	(2) SD	(3) Mean	(4) Min	(5) Max
County–year level variables:					
Exposure to 3G/4G networks	13,955	0.792	1.513	0.162	3.115
Population density	13,955	0.116	0.151	0.000	0.712
Log (population)	13,955	1.452	4.294	0.000	8.129
Share of primary sector	13,955	0.138	0.442	0.005	0.899
Share of secondary sector	13,955	0.369	0.378	0.000	4.416
County–industry–year level variables:					
Firm entry	244,745	461.226	79.187	0.000	39,620.000
$\sinh^{-1}$(firm entry)	244,745	1.982	2.954	0.000	11.280
Log (firm entry)	210,949	1.769	2.694	0.000	10.587
Log (avg. registered capital)	210,949	1.223	5.829	0.000	9.210
Outsider ratio	110,805	0.353	0.511	0.000	1.000

Note: "Exposure to 3G/4G networks" is measured by interacting the inverse of distance to other cities with the logarithm of their annual stock of 3G/4G towers. This interaction is then aggregated across all cities located at least 100 km away from the focal county *i*. "Population density" is defined as the number of people per 1000 square kilometers; "Log (population)" is the logarithm of residential population, measured in units of 10,000 people; "Share of primary (secondary) sector" represents the ratio of the primary (secondary) sector's value-added to the overall gross domestic product. Variables for rural counties are from *China County Statistical Yearbooks* (2013–2017). Variables for urban districts, measured at the city level, are from *China City Statistical Yearbooks* (2013–2017).
"Firm entry" is the number of newly registered private firms at the county–industry level from 2013 to 2017. County–industries with zero firm entry for all 5 years are excluded. $\sinh^{-1}$(firm entry) is the inverse hyperbolic sine of firm entry to address issues with zero entries in some cases, where $\sinh^{-1}(y) = \log(y + \sqrt{y^2+1})$. "Log (firm entry)" is the logarithm of the firm entry variable. "Log (average of registered capital)" is the logarithm of average registered capital (in 10,000 yuan) of newly registered private firms at the county–industry level. "Outsider ratio" is calculated as the ratio of new firms established by nonlocal entrepreneurs to the total number of new firms at the county–industry level from 2013 to 2015. This variable has fewer observations than other county–industry level variables because the information was not available for 2016–2017.
Industry is defined as the one-digit National Economy Industrial Classification (NEIC) code. Firm entry and registered capital were calculated by the authors based on the business registry database.

of zero, we apply an inverse hyperbolic sine transformation to the firm entry variable, that is, $\sinh^{-1}(y) = \log(y + \sqrt{y^2+1})$. This transformation ensures nonmissing values for cells with zero firm entries and closely approximates $log(2y)$ when y is large. The average value of firm entry and $\sinh^{-1}(firm\ entry)$ is 79.2 and 2.95, respectively. For nonzero cells, we directly take the natural logarithm for firm entry and average registered capital, with the mean value as 2.69 and 5.83, respectively. Because hometown information is only available for the entrepreneurs of new firms established in 2013–2015, the number of observations for the "outsider ratio" variable drops to 110,805. On average, nonlocal individuals establish 51% of new firms.

3.2 Empirical estimations

Our main estimating equation is:

$$Y_{ist} = a_1 \cdot Exposure_{it} + a_2 \cdot X_{it} + \delta_{is} + \gamma_t + \varepsilon_{ist} \tag{2}$$

where Y_{ist} represents the total number of newly registered firms in county i, industry s, and year t. $Exposure_{it}$ is the county-level index measuring exposure to 3G/4G networks in county i at year t. X_{it} is a set of control variables, including population density, log($population$), and the share of primary and secondary sectors. δ_{is} represents county–industry-specific fixed effects, accounting for any unobservable factors that remain constant over time within a

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Table 5. Regressions of firm entry on exposure to 3G/4G networks.

	(1)	(2)	(3)	(4)
Sample	**Including zero cells**		**Excluding zero cells**	
Dependent variable:	**$\sinh^{-1}$(firm entry)**		**Log (firm entry)**	
Exposure to 3G/4G networks	0.195***	0.190***	0.196***	0.188***
	(0.016)	(0.016)	(0.017)	(0.017)
Population density		0.178***		0.177***
		(0.066)		(0.065)
Log (population)		−0.013		−0.019
		(0.012)		(0.012)
Share of primary sector		−0.214*		−0.349***
		(0.127)		(0.132)
Share of secondary sector		0.167**		0.134*
		(0.068)		(0.072)
Observations	244,745	244,745	210,949	210,949
County–sector fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes

Note: The sample is restricted to county—one-digit industry—year cells that have at least one firm entry from 2013 to 2017. Standard errors clustered at the county level are reported in parentheses. The symbols *, **, and *** stand for significance levels at the 10%, 5%, and 1% level, respectively.

specific county–industry pair. γ_t represents year-specific fixed effects. ε_{ist} is the error term, which is clustered at the county level. The sample period t spans from 2013 to 2017.

To account for zero values in firm entry, we use both the inverse hyperbolic sine transformation $\sinh^{-1}(firm\ entry)$ and, as a robustness check, the natural logarithm of firm entry for a subsample containing only strictly positive firm entry cells.

3.3 Empirical estimations on firm entry

Having introduced the empirical strategy, in this subsection, we report the regression results on firm entry.

Table 5 presents the estimations of 3G/4G network exposure on the number of firm entries at the county—industry—year level from 2013 to 2017. The sample includes county-industry cells that had at least one firm entry over this 5-year period. Columns (1) and (2) present results using the inverse hyperbolic sine of firm entry and include observations where firm entry is zero. Columns (3) and (4), on the other hand, focus only on cells with positive firm entry, using the natural logarithm of firm entry. Across all these specifications, the coefficients are consistently positive and statistically significant, indicating that greater exposure to 3G/4G networks substantially boosts firm entry. Specifically, based on Column (4), a 1-SD increase in 3G/4G network exposure is associated with a 14.9% (calculated as 0.79*18.8%) increase in firm entry.

There is a potential concern about endogeneity. While the exposure to 3G/4G networks is only correlated with construction behavior of cities from 100 km away, the efficiency of 3G constructions by local governments may not be entirely random. Unobserved factors, such as regional financial capacity, could influence both construction decisions and subsequent firm entries. To address this, we conduct a falsification test by replacing the dependent variable with the number of new firms established between 2008 and 2012. This period had minimal e-commerce activity, yet other regional factors, like financial capacity, would likely still have been influential. If the 3G/4G network exposure was truly correlated with omitted

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Table 6. Falsification test: firm entry in 2008–2012 and exposure to 3G/4G networks.

	(1)	(2)	(3)	(4)
Sample:	**Including zero cells**		**Excluding zero cells**	
Dependent variable:	**$\sinh^{-1}$(firm entry)**		**Log (firm entry)**	
Exposure to 3G/4G networks	0.015	0.013	0.025*	0.020
	(0.013)	(0.013)	(0.014)	(0.014)
Population density		0.203***		0.184***
		(0.057)		(0.055)
Log (population)		0.020		0.024*
		(0.013)		(0.013)
Share of primary sector		0.571***		0.559***
		(0.124)		(0.133)
Share of secondary sector		0.467***		0.522***
		(0.072)		(0.078)
Observations	228,605	228,605	175,141	175,141
County–sector fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes

Note: The sample is restricted to county–one-digit industry–year cells that have at least one firm entry from 2008 to 2012. Firm entry is the number of newly registered private firms at the county–industry level from 2008 to 2012. Standard errors clustered at the county level are reported in parentheses. The symbols * and *** stand for significance levels at the 10% and 1% level, respectively.

variables, we would expect to see a positive and significant coefficient for the exposure index in this falsification test as well. Table 6 presents the results of the falsification test; the exposure variable is not significantly associated with firm entry during this earlier period, supporting the quasi-exogeneity of our exposure variable. Although a small, marginally significant coefficient appears in Column (3), it loses statistical significance once controls are added in Column (4).

Building on the discussion in Section 2.2 about unregistered online businesses, we conduct a robustness check by including self-employed businesses in our measure of firm entry. The data on the number of self-employed businesses comes from the Self-employed Business Registry Database maintained by SAIC (State Administration for Industry and Commerce). Appendix Table A2 further reports results based on a measure of firm entry that combines companies and self-employed businesses. Columns (1)–(2) present estimates using the full sample (including zero-entry cells), while Columns (3)–(4) restrict the sample to cells with positive firm entry. The coefficient for 3G/4G network exposure in this new regression is 0.231 for the full sample and 0.244 for the sample excluding zero cells. These figures are higher than the coefficients reported in Table 5, our main analysis. This finding suggests that the impact of 3G/4G network exposure on firm entry is likely even greater than what our primary analysis captured. The higher coefficient with the broader firm entry measure indicates that our original estimates might have underestimated the true effect by not accounting for self-employed or unregistered businesses.

While acknowledging the limitations of relying solely on registered companies, using this measure may offer two key advantages. First, registered companies provide data on their registered capital, allowing us to see how e-commerce affects the initial registered capital of new businesses-information that is unavailable for self-employed or unregistered entities. Second, for companies registered before 2016, we have access to the hometown information of listed personnel. This unique feature allows us to investigate how e-commerce attracts entrepreneurs from outside the local area, potentially offering insights into the geographical mobility

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Table 7. Regressions of average registered capital and nonlocal fraction of new firms on exposure to 3G/4G networks.

	(1)	(2)	(3)	(4)
Sample:	**2013–2017**		**2013–2015**	
Dependent variable:	**Log (average registered capital)**		**Nonlocal entre. ratio**	
Exposure to 3G/4G networks	−0.076***	−0.074***	0.022***	0.022***
	(0.019)	(0.019)	(0.007)	(0.007)
Population density		−0.216***		0.024
		(0.070)		(0.035)
Log (population)		0.050***		0.004
		(0.013)		(0.007)
Share of primary sector		0.445***		−0.007
		(0.161)		(0.074)
Share of secondary sector		0.219**		0.021
		(0.090)		(0.038)
Observations	210,949	210,949	110,805	110,805
County–sector fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes

Note: The sample is restricted to county–one-digit sector–year cells that have at least one firm entry. The variable "nonlocal entre. ratio" is calculated as the proportion of new firms established by entrepreneurs whose birthplace differs from the county where the firm is registered. Since residency information is no longer available in 2016 and 2017, the data from these two years are excluded in this table. Standard errors clustered at the county level are reported in parentheses. The symbols ** and *** stand for significance levels at the 5% and 1% level, respectively.

linked with e-commerce opportunities. Consequently, we have prioritized the registered company data for our main analysis. Results based on the combined measure are presented in the appendix. This approach aligns with Levine and Rubinstein (2017), who also define entrepreneurship by newly incorporated enterprises.

4. UNDERLYING MECHANISMS AND ENTREPRENEURSHIP COMPOSITION

4.1 Impact of e-commerce on firm startup capital

To check the underlying mechanism, we first examine if e-commerce lowers the threshold for starting a business using the initial registered capital as a measure of startup capital. Although a firm's initial registered capital is not identical to the wealth of shareholders, it still effectively reflects the amount of initial investment by shareholders. In a developing country with an imperfect financial market, the amount of investment to start a business is highly associated with the wealth level of shareholders (Banerjee and Newman 1993; King and Levine 1993). Lower registered capital implies that more potential entrepreneurs can afford to start a business.

Table 7 (Columns 1–2) reports the findings about the relationship between 3G/4G network exposure and the log of average startup capital of newly registered firms. We exclude cells with zero firm entry since registered capital information is not applicable in those cases. Essentially, we examine the impact on the intensive margin of registered capital. The control variables and fixed effects are consistent with those in Table 5. The coefficients for the 3G/4G network exposure are negative regardless of having control variables or not. According to Column (2), a 1-SD increase in the 3G/4G exposure is associated with a 5.85% (calculated as 0.79*7.4%) decrease in the average registered capital of new firms. This represents a sizable economic effect.

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4.2 Role of e-commerce in reducing reliance on social networks

In traditional economies reliant on personal trade, locals typically have an advantage, possessing better access to local social networks and financial resources than nonlocals (Banerjee and Munshi 2004; Michelacci and Silva 2007). We hypothesize that e-commerce, with its emphasis on timely payments and impersonal transactions, diminishes the importance of local social networks. This shift could level the playing field, offering nonlocals an equal opportunity to enter and succeed in related businesses. Our hypothesis, therefore, predicts that e-commerce disproportionately benefits nonlocals.

We infer residency information from the first six digits of the resident identification cards (IDs) of legal representatives, which are provided at the time of registration. However, the registration data available to us only records the first six digits of the IDs for entrepreneurs who registered before 2016. Thus, we have to use only a subsample in the period 2013–2015 to test whether e-commerce can reduce the need for social networks to start a business. We measure the reliance on social networks using the fraction of new firms established by nonlocal entrepreneurs. Nonlocal entrepreneurs are defined as legal representatives whose birth counties differ from their firms' registered counties, calculated at the county-1-digit industry–year level.

To test the hypothesis, we regress the fraction of nonlocal firm entries on the 3G/4G network exposure. Columns (3) and (4) of Table 7 presents these estimations, both with and without control variables. The control variables and fixed effects are still consistent with those in Table 5. The coefficients for the 3G/4G network exposure are consistently positive and statistically significant. According to Column (4), a one-standard-deviation increase in the 3G/4G network exposure is associated with a 3.4% (calculated as 0.79 * 2.2%/0.51) rise in the nonlocal ratio.

4.3 E-commerce effect on different type of firms

Building on the insight from the Baigou case study on the varying effects of e-commerce expansion on different businesses, this section examines two distinct categories: firms directly involved in e-commerce (e-commerce related) and those operating exclusively offline (non-e-commerce).

However, the 4-digit industry codes assigned by the business registry may not always precisely reflect the actual business types. For example, the "manufacturing sector" does not have a specific code for companies that both produce goods and directly sell them to online retailers. To overcome this limitation, we leverage natural language processing (NLP) techniques, similar to the method by Cong et al. (2024). This approach analyzes the "business operation scope" field in the business registry data to accurately identify enterprises engaged in e-commerce activities, as demonstrated by Cong et al. (2024).

To understand the subtle effects of e-commerce on different business types, we categorize firms within the manufacturing and wholesale/retail sectors. Based on the interaction between industry and e-commerce activity, we create four distinct groups: e-commerce-related manufacturing companies, non-e-commerce manufacturing companies (traditional producers), e-commerce merchants, and non-e-commerce merchants (without their manufacturing facility). Figure 5 illustrates the relationship between 3G/4G network exposure and firm entries for these four categories. It is apparent that 3G/4G network exposure is positively associated with the entries of e-commerce related manufacturers and merchants. Conversely, traditional producers and non-e-commerce merchants do not see a significant increase in entries with e-commerce growth. Interestingly, e-commerce development even appears to have a crowding-out effect on the entry of traditional merchants.

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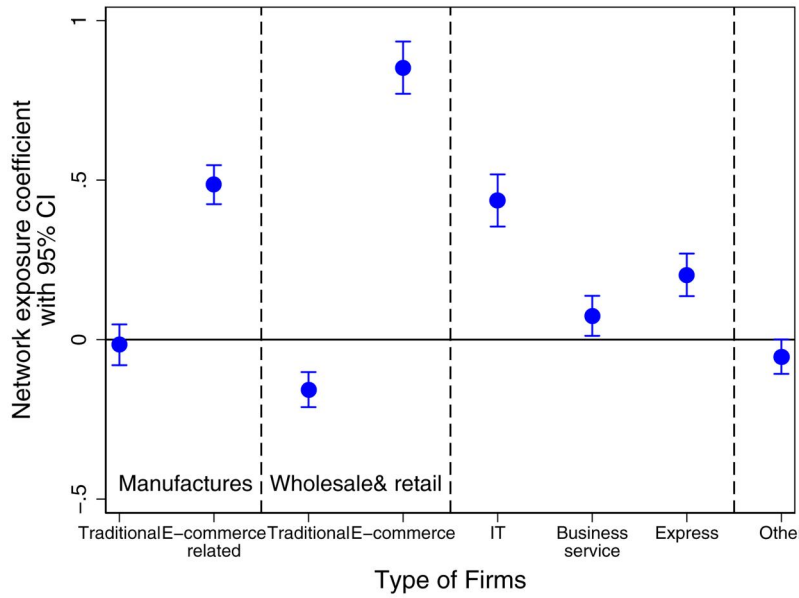


Figure 5. The heterogeneous effect of exposure to 3G/4G networks on firm entry. *Source:* We analyze the relationship between e-commerce development and firm entry at the county level in China based on business registry data. To identify e-commerce-related businesses, we adopt the same Natural Language Processing (NLP) approach utilized by Cong et al. (2024) on the "business scope" text field within the registry data. Within the manufacturing sector, firms are categorized as either traditional manufacturers or e-commerce-related manufacturers based on the NLP analysis. Similarly, in the wholesale and retail sector, firms are classified as traditional merchants or e-commerce merchants. Firms in the IT, business services, and express delivery sectors are classified based on their two-digit industry codes. All remaining firms are grouped into the "other industries" category. A 95% confidence interval, adjusted for county-level clustering at the county level, is plotted as vertical lines.

We further investigate firm entry in three sectors crucial for online retailers in Section 2.4: IT services, business services, and express delivery. All remaining firms are categorized as "other industries." The results, presented in Figure 5, show that the 3G/4G network exposure has a significantly positive coefficient for all three service sectors. In contrast, "other industries," which lack direct connections to e-commerce, do not show this significant positive association.

4.4 Composition of Entrepreneurship

To understand if the evolving nature of entrepreneurship is tied to entrepreneurs' personal characteristics like education and experience, we turn to the ESIEC data. This firm-level survey uses stratified sampling from the Business Registry Database and covers private firms in six Chinese provinces (Henan, Guangdong, Zhejiang, Shanghai, Gansu, and Liaoning). The 2018 wave interviewed 6198 firms, including 3185 founders of corporate companies (the rest were self-employed business owners or corporate managers). Our analysis focuses on 2643 corporate firms where the entrepreneur, not a manager, was successfully interviewed and complete data was available.[10]

[10] The final sample contains 2310 observations because singleton groups are automatically dropped when controlling for county-sector fixed effects and year fixed effects.

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Table 8. Regressions of entrepreneur demographics on 3G/4G network exposure.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Dependent variable:	**Education**		**Employee experience**		**Experience outside the location**		**Age below 35**		**Female**	
3G/4G network exposure	0.168	0.190*	0.267**	0.211*	0.239**	0.149	0.164	0.004	0.033	0.044
	(0.112)	(0.109)	(0.121)	(0.112)	(0.108)	(0.109)	(0.109)	(0.114)	(0.091)	(0.093)
Population density		0.265		0.041		−0.077		−0.083		−0.072
		(0.178)		(0.083)		(0.135)		(0.138)		(0.209)
Log (population)		−0.035		−0.010		0.029		0.040		0.100*
		(0.087)		(0.048)		(0.056)		(0.057)		(0.054)
Share of primary sector		−0.960		−0.998*		−0.761		−1.801**		0.564
		(0.670)		(0.574)		(0.562)		(0.732)		(0.506)
Share of secondary sector		−0.981**		−0.379		0.063		−0.314		0.197
		(0.452)		(0.351)		(0.379)		(0.543)		(0.339)
Observations	2310	2310	2310	2310	2310	2310	2310	2310	2310	2310
Mean of D.V.	0.530	0.530	0.778	0.778	0.646	0.646	0.502	0.502	0.212	0.212

Note: The sample is drawn from the ESIEC firm survey, which is a random sample based on the Business Registry Database. The analysis includes new firms established between 2013 and 2017. County–sector fixed effects and year fixed effects are controlled in all specifications. "Education" is a dummy variable, equal to one if the respondent's education level is above middle school. "Employee experience" is a dummy variable, equals one if the respondent has ever worked as an employee. "Experience outside the location" is a dummy variable, equal to one if the respondent either worked in or was born in a place different from the current firm's location. "Age below 35" is a dummy variable, equal to one if the respondent founded the firm before the age of 35. "Female" is a dummy variable indicating the respondent's gender. Standard errors clustered at the county level are reported in parentheses. The symbols * and ** stand for significance levels at the 10% and 5% level, respectively.

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Using this dataset, we have constructed a panel of entrepreneur cohorts based on their firm's establishment year and matched them to county–year 3G/4G network exposure data. We then replicate the baseline empirical specification at the entrepreneur level, incorporating two sets of outcome variables. The first captures demographic characteristics: education (a dummy for above middle school), prior employment experience, age, gender, and outside-work experience. The second measures professional background. Following Levine and Rubinstein (2017), we infer managerial experience from prior occupation and categorize the last employment into state-related (government, public institutions, and state-owned enterprises (SOEs)), private, and foreign-owned sectors.

As Table 8 and Appendix Table A3 demonstrate, increased access to e-commerce is linked to entrepreneurs being more likely to have completed high school or higher education and possessing prior work experience, particularly from outside their home region. Our findings reveal no significant relationship with managerial occupational experience or experience within the state or private sector.

However, there's weak evidence of a positive association with foreign sector experience. This suggests that successful e-commerce entrepreneurs might benefit more from a broader market awareness than from traditional managerial skills. It is important to note that these quantitative analyses control for county–sector and year-fixed effects, essentially focusing on within-sector differences. This may explain why our findings, based on quantitative analysis, diverge slightly from Baigou's qualitative case study, which compares traditional and e-commerce sectors.

Overall, these analyses offer a comprehensive understanding of e-commerce's impact on the makeup of entrepreneurs across both extensive and intensive margins. Our findings indicate that e-commerce reshapes the composition of entrepreneurs along two dimensions. Across sectors, we observe more substantial increases in firm entry in e-commerce-related sectors compared to traditional sectors. Within sectors, there is a noticeable shift toward more educated, externally experienced entrepreneurs who tend to exhibit reduced reliance on local social capital.

5. CONCLUSION

History shows that the transition from personal trade to impersonal trade takes a long time. Sound institutions are the foundations for impersonal trade. However, it is a daunting task to build sound institutions in the first place. In this article, we use China as an example to show how e-commerce has offered a faster route for the transition to impersonal trade.

E-commerce has effectively lowered capital entry barriers and reduced the reliance on social networks in business. This creates a new opportunity for aspiring entrepreneurs who were previously limited by insufficient financial and social capital. Drawing insight from a case study and leveraging the administrative universe business registry database as well as firm survey data, our article shows that e-commerce has particularly benefited nonlocal entrepreneurs, thereby unlocking previously inhibited entrepreneurial talent. E-commerce has also created new business opportunities within supporting industries, such as courier services, IT, designers, and marketing agencies. Furthermore, within existing sectors, we are observing a noticeable shift toward a more educated and externally experienced form of entrepreneurship as e-commerce continues to expand.

Our findings have important policy implications. An increasing number of countries, including China, have implemented regulations on e-commerce platforms. When formulating the regulations, it is necessary to consider the positive effect of e-commerce in unleashing entrepreneurial activities.

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AUTHOR CONTRIBUTIONS

Ruochen Dai and Xiaobo Zhang contributed equally to this work. Ruochen Dai and Xiaobo Zhang conducted the investigation, collected and curated the data, conceived the study, performed the formal analysis, and wrote and reviewed the manuscript.

Table A1. E-commerce development index and 3G/4G network exposure.

	(1)	(2)	(3)	(4)
Dependent variable:	**Log (online business index + 1)**		**Log (online shopping index + 1)**	
Exposure to 3G/4G networks	0.125***	0.123***	0.001	0.000
	(0.025)	(0.025)	(0.002)	(0.002)
Population density		−0.055		−0.024***
		(0.044)		(0.008)
Log (population)		−0.038**		0.001
		(0.018)		(0.001)
Share of primary sector		−0.243		−0.099***
		(0.261)		(0.020)
Share of secondary sector		0.007		0.014
		(0.121)		(0.010)
Observations	13,955	13,955	13,955	13,955
County fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes

Note: The sample is at the county–year level. The OBI, developed and published annually by Alibaba, measures a county's capacity for online commerce. It is constructed using two components: the number of online retailers per capita and the share of large online retailers (those with annual transaction volumes exceeding RMB 240,000). The OBI primarily reflects a county's ability to sell products online. In contrast, the online shopping index (OSI) captures online consumer activity. It is based on the number of online buyers per capita and the share of buyers with annual online consumption exceeding RMB 10,000. Standard errors clustered at the county level are reported in parentheses. The symbols ** and *** stand for significance levels at the 5% and 1% level, respectively.

Table A2. Robustness analysis: including self-employed businesses in the sample.

	(1)	(2)	(3)	(4)
Sample	**Including zero cells**		**Excluding zero cells**	
Dependent variable:	**$\sinh^{-1}$(firm entry)**		**Log (firm entry)**	
Exposure to 3G/4G networks	0.233***	0.231***	0.248***	0.244***
	(0.017)	(0.017)	(0.017)	(0.017)
Population density		0.297***		0.309***
		(0.087)		(0.089)
Log (population)		−0.009		−0.012
		(0.012)		(0.012)
Share of primary sector		0.088		−0.017
		(0.135)		(0.137)
Share of secondary sector		0.243***		0.196***
		(0.073)		(0.075)
Observations	248,650	248,650	222,151	222,151
County–sector fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes

Note: The sample is restricted to county–1-digit–industry cells with at least one company or self-employed business registered between 2013 and 2017. The number of self-employed businesses is from the Self-employed Business Registry Database. Standard errors clustered at the county level are reported in parentheses. The symbols *** stands for significance levels at the 1% level.

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Table A3. Entrepreneur's experience and exposure to 3G/4G networks.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable:	**Manager experience**		**State sector experience**		**Private sector experience**		**Foreign sector experience**	
3G/4G network exposure	0.089	0.094	0.142	0.169	0.128	0.068	0.074*	0.032
	(0.076)	(0.088)	(0.100)	(0.110)	(0.118)	(0.134)	(0.041)	(0.054)
Population density		0.132		0.254***		−0.110		0.036
		(0.154)		(0.067)		(0.154)		(0.063)
Log (population)		0.025		−0.014		0.006		0.007
		(0.045)		(0.040)		(0.047)		(0.026)
Share of primary sector		−0.567		−0.263		−0.672		−0.293
		(0.483)		(0.528)		(0.764)		(0.317)
Share of secondary sector		−0.621		−0.339		−0.158		0.169
		(0.403)		(0.339)		(0.475)		(0.277)
Observations	2310	2310	2310	2310	2310	2310	2310	2310
Mean of D.V.	0.178	0.178	0.230	0.230	0.517	0.517	0.0844	0.0844

Note: The sample is drawn from the ESIEC firm survey, a random sample based on the Business Registry Database. The analysis includes new firms established between 2013 and 2017. County–sector fixed effects and Year fixed effects are controlled in all specifications. "Manager experience" is a dummy variable, equal to one if the respondent has ever held a managerial position. "State sector experience" is a dummy variable, equal to one if the respondent ever worked in the state sector, including government agencies, public institutions, or state-owned enterprises. "Private sector experience" is a dummy variable, equal to one if the respondent has worked in privately owned companies. "Foreign sector experience" is a dummy variable, equal to one if the respondent has worked in foreign-owned firms. Standard errors clustered at the county level are reported in parentheses. The symbols * and *** stand for significance levels at the 10% and 1% level, respectively.

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DATA AVAILABILITY

The data underlying this article will be shared on reasonable request to the corresponding author. Due to non-disclosure agreements and confidentiality requirements, the raw data cannot be

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publicly released. Secure access for replication is available through the Center for Enterprise Research (CER), Peking University (https://cer.gsm.pku.edu.cn/about/data.htm).

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